Detecting Seizure Onset Patterns as Temporal Objects

A Generalizable YOLO-Based Framework for iEEG Time Series Event Detection

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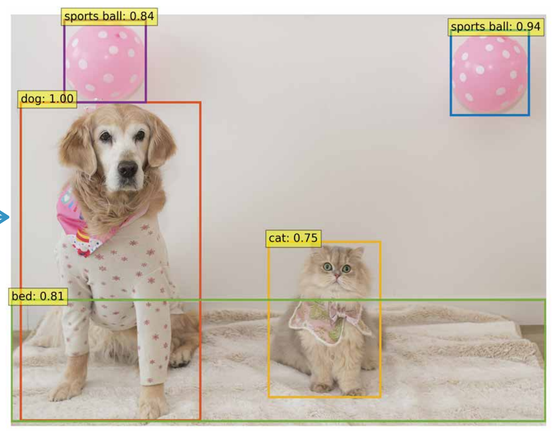
Background

- SOP characteristics help localize the seizure onset zone (SOZ)
- In MTLE, SOP type predicts post-surgical outcome
- Expert annotation is time-consuming, subject to inter-rater variability, and bottlenecks population-scale analysis
- Existing automated tools detect seizure onsets but do not classify complex SOP morphology

Engineering Challenge

- EEG events span 3 orders of magnitude: spikes (~300 ms) to seizures (>60 s)
- Multiple event classes co-occur and overlap within and across channels
- High inter-patient morphological variability

Inspired by:



Object detection natively handles multi-scale, overlapping events

We present the first deep learning pipeline for automated, multi-class, per-channel SOP detection, generalizable to other ictal and interictal EEG patterns.

Method

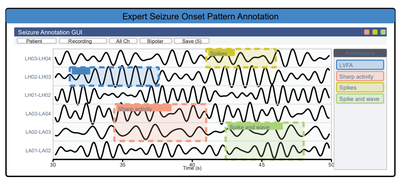
Patient Data

57 Patients
30F / 27M
Age: 20–70
Surgery: 49, Other: 8
Focal: 48, General: 9
Favorable (Engel@12m): 44

Expert SOP Annotation

Expert annotators draw *per-channel* bounding boxes via custom GUI using standardized labels

LVFA, Rhythmic, Sharp, Spike



Data Processing

Dataset

57 Patients
143 Seizures
1549 Annotations

Preprocessing

Bandpass Filter
Bipolar Montage
60s Sliding Windows

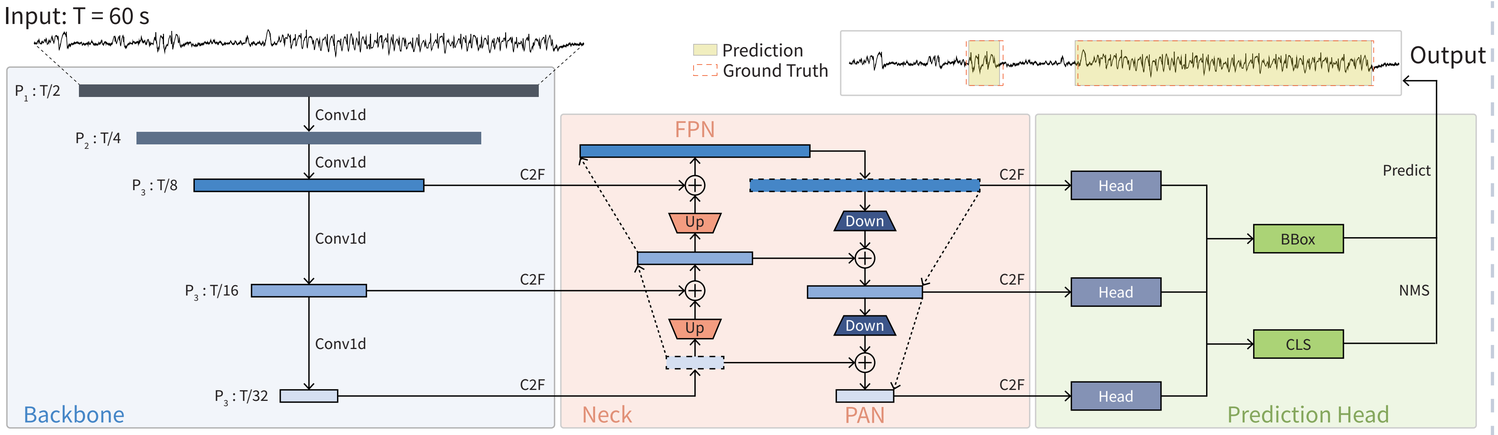
Model Training

Backbone — Feature extraction → Neck — FPN / PAN fusion → Head — Predict bbox + cls → LVFA, Spikes, S&W

Goal: learn to replicate expert SOP annotations

YOLOv8 1D

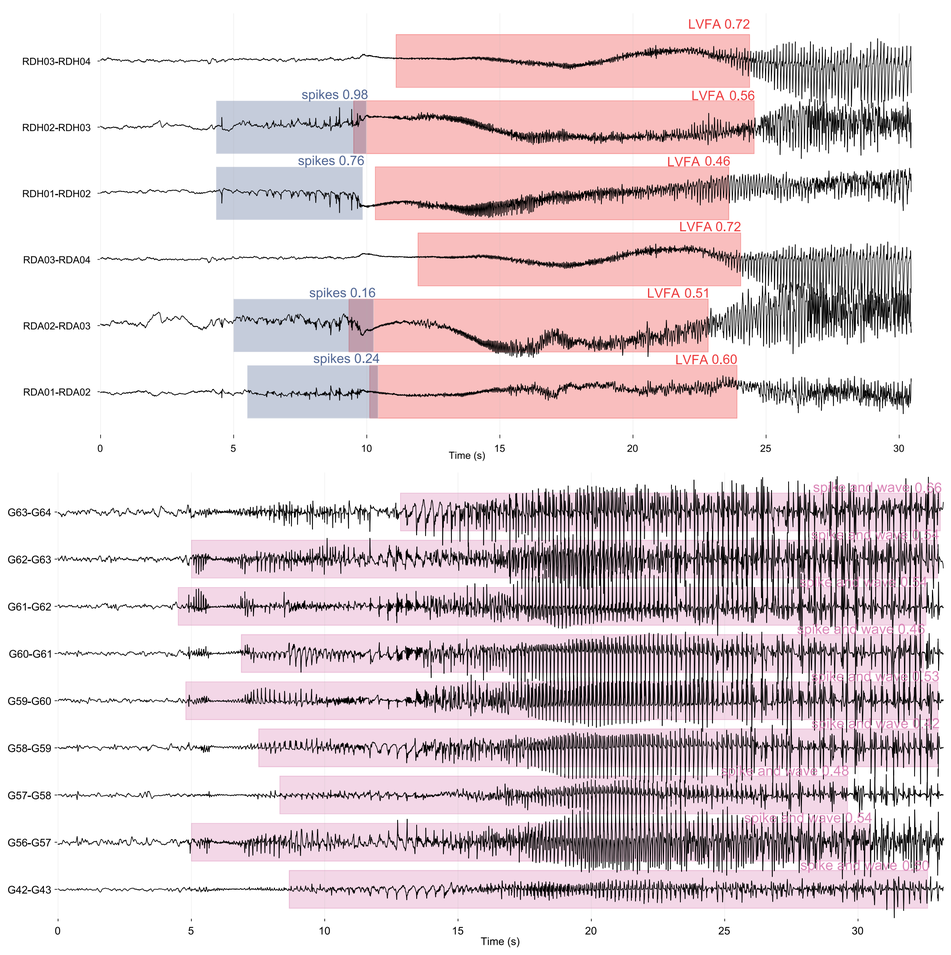
YOLO (**Y**ou **O**nly **L**ook **O**nce): Encodes features at multiple scales and **simultaneously** predicts *class*, *onset*, and *offset* in one pass.



2D convolutions replaced with 1D temporal equivalents | Anchor scales adapted to EEG event durations | Localization loss and NMS adapted for 1D IoU

Results

SOP Detection: Example Prediction



Experiments

Exp 1 - SOP Detection

Main result - no direct benchmark

- Leave-patient-out 5-fold CV
- Inter-rater benchmark on independent dataset

Exp 2 - Generalizability

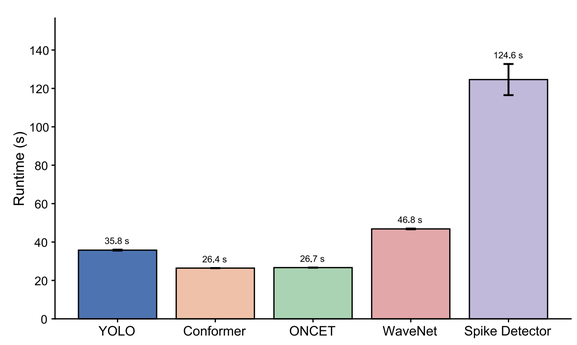
Benchmark other than SOP

- Joint spike & seizure detection
- Benchmark against single task SOTA model on performance and inference speed

Cross-task generalization without performance loss

Model	% Precision	% Recall	mAP@50	mAP@50-95
Ours - spike & seizure	0.579 ± 0.002	0.648 ± 0.002	0.652 ± 0.001	0.526 ± 0.001
Spike Detector	**0.712 ± 0.013**	0.575 ± 0.013	n/a	n/a
Ours - spike only	0.576 ± 0.003	**0.650 ± 0.002**	0.614 ± 0.002	0.449 ± 0.002
WaveNet	0.507 +- 0.096	**0.660 ± 0.024**	n/a	n/a
ONCET	0.526 ± 0.001	0.625 ± 0.002	n/a	n/a
Ours - seizure only	**0.596 ± 0.002**	0.640 ± 0.002	0.691 ± 0.001	0.603 ± 0.002

Multi-class detection at single-task detector speed

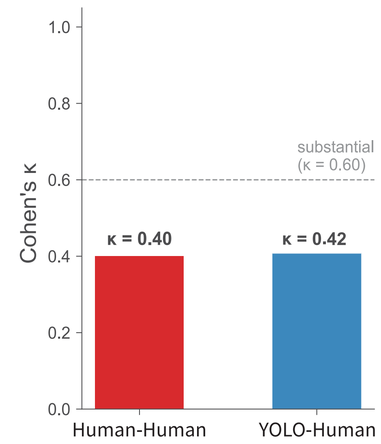


Model	Type	Device
YOLO	CNN	GPU
Conformer	Transformer	GPU
ONCET	CNN	GPU
WaveNet	CNN	GPU
Spike Detector	Template based	CPU

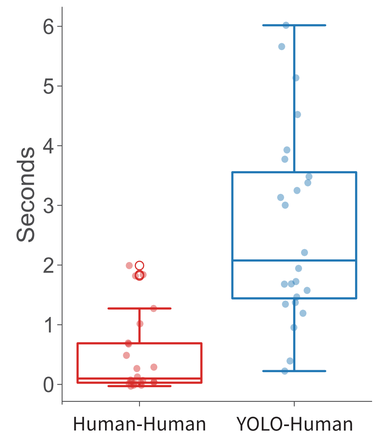
Robust detection across leave-patient-out cross-validation

Model	% Precision	% Recall	mAP@50	mAP@50-95
YOLOv8-1D (OURS)	0.635 ± 0.027	0.848 ± 0.022	0.683 ± 0.017	0.603 ± 0.021

SOP class prediction within inter-rater agreement



Onset timing approaches inter-rater precision



Conclusion & Future Directions

- First automated pipeline for multi-class, per-channel SOP detection in iEEG
- SOP class agreement within expert inter-rater range
- Inference speed competitive with specialized single-task detectors
- Generalizes to other EEG event types without task-specific redesign
- Expand training data and label diversity to improve detection robustness
- Extend to additional EEG patterns and clinical applications
- Open-source code, model weights, and annotated dataset upon publication

Funding Source — NIH National Institutes of Health

Stay Tuned — GitHub